Tyler Lacy

Electrical Engineering Student · NASA Lunabotics Systems Engineering Lead · U.S. Army Veteran

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SUMMARY

Electrical Engineering student, U.S. Army veteran (Secret security clearance), and award-winning NASA Lunabotics systems engineering lead. Combines hands-on electro-mechanical tool reliability, high-pressure fluid testing, and work-order/inventory management experience with C++, Python, and Linux embedded systems development. Proven track record driving campus STEM engagement and leading multidisciplinary teams through systems engineering milestones (PDR/CDR). Seeking a research-focused engineering internship.

EDUCATION

Georgia Institute of Technology — B.S. Electrical Engineering, incoming transfer student (2030).

Lone Star College — Associate of Science, Engineering Track, Honors College, expected Spring 2027. Relevant coursework: Programming Fundamentals I & II (C++), General Chemistry I, University Physics I, Calculus II, Macroeconomics.

TECHNICAL SKILLS

Programming & Systems: C++, Python, Linux (Ubuntu/Pop!_OS), Git/GitHub, ROS 2, Bash, Embedded C.

Hardware & Electronics: ESP32, LoRa (915 MHz), Raspberry Pi, Jetson Orin Nano, Arduino/Teensy, LiDAR, I2C/SPI/UART protocols.

Design & Prototyping: Autodesk Fusion 360 (CAD/CAM), OnShape, 3D printing (FDM/PLA+/PETG-CF), manual/CNC machining, calipers, micrometers, GD&T basics.

Testing & QA: Hydrostatic & pneumatic pressure testing, seal/thread redress, work order tracking (ERP/SAP), 5S, lean QA/QC.

PROFESSIONAL EXPERIENCE

R&M Technician III — Weatherford International (Odessa, TX · 2022–2024) — Inspected, redressed, and hydrostatically pressure-tested high-pressure liner-hanger running tools to ensure zero-defect redeployment, generating digital work orders and managing staging inventory alongside rig operators to reduce non-productive time. Streamlined warehouse parts allocation and improved 5S throughput, helping the team build 2–6 fully certified cement heads per day.

RESEARCH & ENGINEERING PROJECTS

Systems Engineering Team Lead — NASA Lunabotics Rover (2025–Present) — Led an award-winning multidisciplinary engineering team through NASA's Systems Requirement Review and Preliminary/Critical Design Reviews for an autonomous regolith excavator, engineering 4-motor drivetrain integration and sensor power distribution across microcontrollers, relay buses, and 2D LiDAR navigation, while formulating risk mitigation matrices and budget allocations exceeding $1,000.

Project Lead — NASA ESP32 IoT Mesh Network (In progress) — Working with the Computer Science Club, architected a 10-node off-grid LoRa/ESP32 mesh network across campus to optimize power consumption, programming low-power C++ firmware with dynamic multi-hop routing and automated data transmission to an edge gateway.

Fluid System Pressure Testing & Tool Reliability — Weatherford International / Independent (2022–2024) — Executed hydrostatic pressure testing protocols on high-pressure fluid tools to ensure leak-free seal integrity prior to field deployment, logging equipment maintenance and certification records through digital work order systems while tracking staging inventory across product lines.

Autonomous Drive Developer — VEX U Robotics Platform (2025) — Developed modular C++ drivetrain control algorithms for a 6-wheel tank-style robot chassis and programmed autonomous routines with tuned PID velocity response curves, competing in multiple competitions among top schools from around the world.

LEADERSHIP & MILITARY SERVICE

25U Signal Support Systems Specialist — U.S. Army National Guard (Secret clearance, inactive) — Operated, deployed, and maintained tactical communications networks, radio systems, wire/fiber networks, and secure encryption hardware, earning the National Defense Service Medal and Army Service Ribbon.

Event Manager & Lead — ASME Chapter, Lone Star College (2025–Present) — Manage chapter build budgets and organize campus outreach events, co-leading Lunabotics robotics initiatives to expand student participation in engineering.

Historian — Computer Science Club, Lone Star College (2025–Present) — Enhanced LSC–CyFair STEM engagement by organizing hands-on technical workshops, maintaining public code archives, and documenting club engineering initiatives.